SENTINEL AI: A CPU-Efficient and Privacy-Conscious Real-Time Indoor Crowd Monitoring and Occupancy Intelligence System Using Computer Vision and Temporal Analysis

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Abstract — Indoor institutional occupancy monitoring is essential for safety regulation, energy efficiency, and crowd management in modern smart campuses. However, prevalent surveillance paradigms rely heavily on cloud-connected infrastructures and privacy-invasive facial recognition, or demand specialized NVIDIA GPU accelerators incompatible with standard commodity institutional hardware. This paper presents SENTINEL AI, a modular, privacy-conscious, and strictly CPU-efficient edge computer vision platform for real-time indoor crowd monitoring. Deployed and evaluated on a Dell Latitude 3420 laptop powered by an 11th Gen Intel Core i5-1145G7 CPU without discrete graphics, the system integrates SSDLite320 with a MobileNetV3 Large backbone for person detection, coupled with an online IoU multi-object tracker, bidirectional virtual tripwire counting, and an image quality reliability layer based on Laplacian blur and photometric metrics. Empirical evaluation demonstrates that the proposed SSDLite320 detector achieves true full-body pedestrian detection at an average latency of 110.98 ms (9.01 FPS) on commodity CPU, outperforming classical Haar cascade baselines. Incorporating temporal multi-object tracking reduces frame-to-frame count variance by 100% under optical dropouts, while virtual tripwire analysis maintains a reconciled occupancy state.

1. Introduction & Problem Statement

Institutional facilities require reliable information regarding occupancy density and movement flux. However, existing commercial systems mandate discrete GPUs or transmit unencrypted video streams to cloud servers. SENTINEL AI demonstrates that commodity quad-core laptop CPUs can achieve robust, sub-150ms real-time person detection and multi-object tracking with zero biometric identification and zero raw video retention.

2. Empirical Benchmarks (Dell Latitude 3420 CPU)

Detector Model	Target Class	Latency (ms)	FPS	Model Size
Haar Baseline	Face & Upper Body	39.91 ms	25.05 FPS	0.93 MB
SSDLite320 MobileNetV3	COCO Person (Full)	110.98 ms	9.01 FPS	13.4 MB

3. Conclusion & Ethical Stance

SENTINEL AI provides an empirically grounded blueprint for privacy-preserving, CPU-efficient edge monitoring in academic campuses. All detection runs locally with ephemeral tracking IDs, proving that modern deep learning crowd intelligence can be successfully deployed on commodity institutional hardware without sacrificing privacy or requiring expensive GPUs.